

Architecture

Component decomposition of the conversational-memory system, and how each component earns its place.

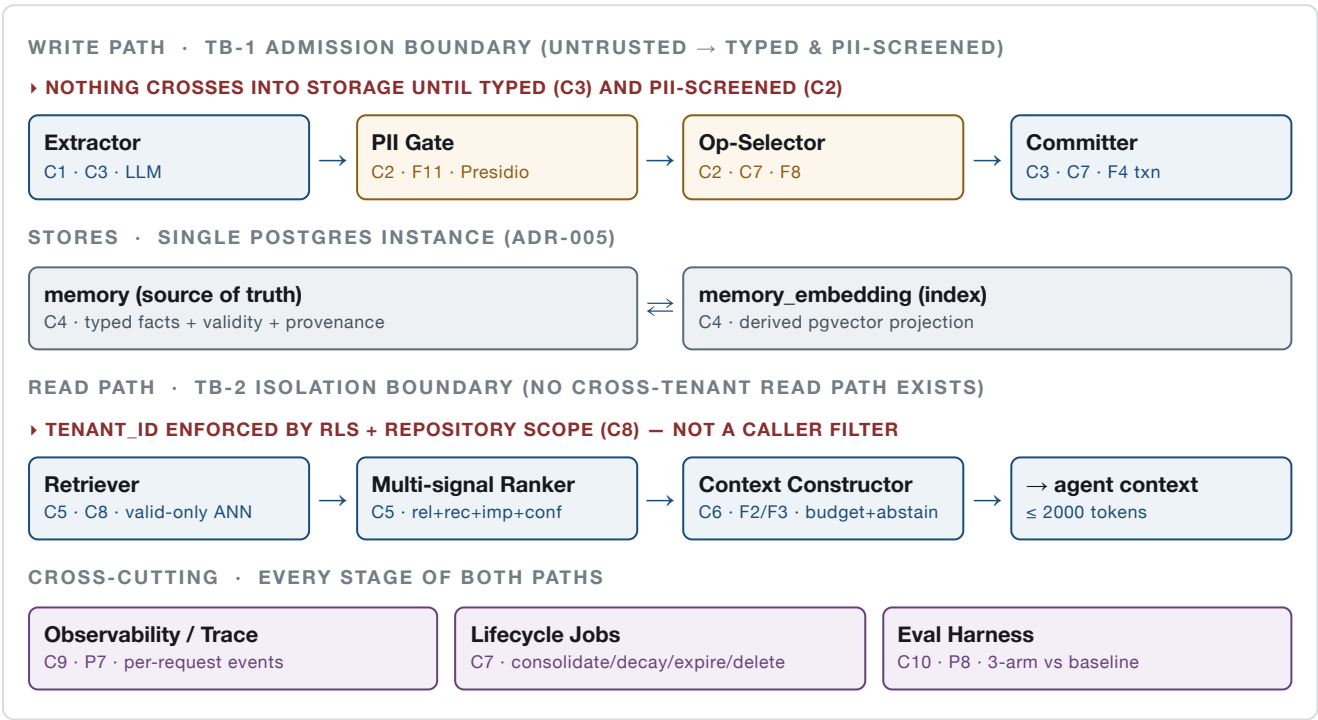
System Conversational Memory Intelligence System

Author Dheeraj Pranav

Companion system_design.pdf · data_flow.pdf

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1. Component diagram



Blue = processing components; grey = stores; amber = admission gates (TB-1); purple = cross-cutting. The ↔ between stores is the C4 projection relation: the index is rebuilt from the source of truth, and deletion cascades across both in one transaction.

2. Why each component exists (traceability)

The handbook completion standard requires that every component correspond to a requirement, invariant, or measured failure. This is that mapping — no orphan components, no unaddressed failures.

COMPONENT	CAPABILITY	FIXES / MEASURED	FORCED BY
Extractor	C1, C3	F1, F7	P1 (nothing persists), P3 (chunks aren't facts)
PII Gate	C2	F11 · D3: 3 exposures	P6 (ineligible data exists; hard gate only)
Op-Selector	C2, C7	F8 · D3: 0.556 wasted	P3 (redundancy is a policy problem)
Committer	C3, C7	F4 · D3: 0.80	P4 (validity + supersession, ADR-001)
memory store	C4	F9	P6, P8 (authoritative + rebuildable)
embedding index	C4	—	P8 (re-projection, not migration)
Retriever	C5, C8	F10 · D3: 7/11 leaked	P5 (isolation invariant, ADR-005)
Ranker	C5	F4, F8	P3, P4 (similarity ≠ validity)
Constructor	C6	F2, F3 · D3: cold-start 2/2	P2 (finite, position-sensitive context)
Observability	C9	all (indirectly)	P7 (decisions invisible in output)
Lifecycle Jobs	C7	F1, F6, F9	P4, P6, P3
Eval Harness	C10	credibility of all	P8 (quality is distributional)

3. The two boundaries that carry the safety posture

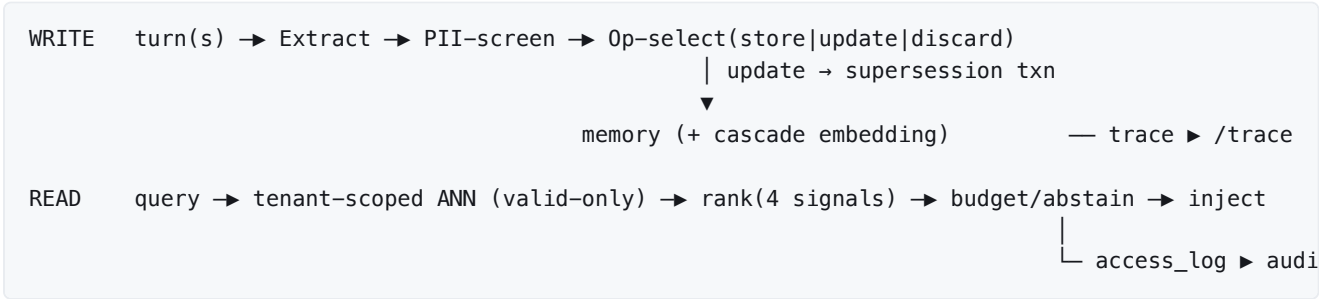
TB-1 — ADMISSION (WRITE PATH)

Untrusted conversation becomes stored state only after extraction (typed, resolved) and a **hard** PII gate. There is deliberately **no** post-storage redaction path: by the time data is on the output it is already in the index and backups (P6). The gate is the only correct place.

TB-2 — ISOLATION (READ PATH)

Stated as "**no query path can express a cross-tenant read**," not "every path filters" (C8). Two enforcement layers (RLS + constructor-scoped repository) mean an application bug is caught by the database and an RLS misconfig is caught by the code shape. D3 measured why this must be structural: the naive filter-mindset leaked on 7/11 queries, and better similarity makes near-duplicate foreign memories *more* retrievable, not less.

4. Sequence sketch (detailed data-flow → data_flow.pdf)



This is the component-level flow; the full request/response sequence diagrams, the op-selector decision tree, and the deletion cascade timing go in `data_flow.pdf` (next artifact). Data structures for each edge are in `data_model.md`; the call contracts in `api_contracts.md`.

AI assistance disclosure. Claude helped render this architecture. The decomposition, the boundary choices, and the traceability mapping are mine and are what I will defend in review.